

Computing and Comparing Basic Performance Measures

Objective:

- Our objective with this exercise is to compute some basic performance measures using SAS, and to examine how our ranking of a group of managers changes under these different measures. Thus, we look at some of the issues pointed out by Roll (1978) and others. The following SAS datasets are available for you to complete today's list of tasks:
 - A. "net_returns_for_crystal_ball"
 - B. "t-bills and s&p 500 (monthly)"
 - C. "msp500"

A. “netreturns”

- This file contains monthly net returns for several U.S. mutual funds.

B. “tbill”

- This file contains the monthly returns for 30-day U.S. Treasury Bills (“tbillretto”) and S&P 500 stocks (“sp500retto”; value-weighted and rebalanced monthly, with cash dividends reinvested).

C. “msp500”

- This file contains the monthly returns for both the equal-weighted (“ewretd”) and value-weighted (“vwretd”) S&P 500 stocks (rebalanced monthly, with cash dividends reinvested).

Hw7 Please complete the following tasks:

1. Compute the arithmetic average monthly return of each fund, and rank the funds by this simple measure.
2. Compute the geometric average monthly return of each fund, and re-rank the funds. How and why is this ranking different from #1?
3. Compute the Jensen measure of each fund, and re-rank the funds. Assume that the value-weighted S&P 500 index is the “market.” How and why is this ranking different from #1?
4. Repeat #3, this time using the equal-weighted S&P 500 index as the “market.” How and why is this ranking different from #3?
5. Compute the tracking error variance of each fund, using the value-weighted S&P 500 as the benchmark. Re-rank funds by this measure (that is, the top fund will have the lowest tracking error variance, and the bottom fund will have the highest TEV). Thus, do not consider the tracking error gain in this exercise. What does this ranking tell you about the risk-taking behavior of these funds? What doesn't it tell you (i.e., systematic vs. idiosyncratic risk-taking)?

Please complete the following tasks (2):

6. Using the Jensen model of #3, compute the information ratio of each fund, and re-rank the funds. How and why is this ranking different from #3?